

CAPITAL STRUCTURE IN PERFECT MARKETS

(optimum, effects on risk and value, MM theorem)

CAPITAL STRUCTURE IN PERFECT MARKETS

(OPTIMUM, EFFECTS ON RISK AND VALUE, MM THEOREM)

Source

- *Corporate Finance*, Jonathan B. Berk and Peter DeMarzo: Ch 14, 15
- *Principles of Corporate Finance*, Brealey, Myers, Allen, Edmans (14th Ed.): Ch 16

Perfect capital markets

1. Investors and firms can *trade the same set of securities at competitive market prices* equal to the present value of their future cash flows.
2. There are *no taxes, transaction costs, or issuance costs* associated with security trading.
3. A firm's *financing decisions do not change the cash flows* generated by its investments, nor do they reveal new information about them.

Capital structure

- The **capital structure** shows the *relative proportions of debt, equity, and other securities* that a firm has outstanding. A firm can be financed by equity only, or it can be financed in part by equity and in other part by debt.
- Let's say there is a firm that takes on a risky project. Because a project depends on the overall economy, it contains market risk, thus investors require a risk premium. As a result, the CF from the project will be discounted by r_f (5%) + risk premium (10%).

TABLE 14.1

The Project Cash Flows

Date 0	Date 1	
	Strong Economy	Weak Economy
-\$800	\$1400	\$900

- If this project/firm is financed using equity alone (the firm has no other liabilities) then equity holders will receive all the cash flows (CF) generated by the project (the CF of equity)

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are the same as the CF of the project) and equity in such firm will be called *unlevered equity*.

- The expected CF is \$1,150, so the MV(E) is $\$1,150/1.15 = \$1,000$:

TABLE 14.2

Cash Flows and Returns for Unlevered Equity

	Date 0	Date 1: Cash Flows		Date 1: Returns	
	Initial Value	Strong Economy	Weak Economy	Strong Economy	Weak Economy
Unlevered equity	\$1000	\$1400	\$900	40%	-10%

- A firm can also be financed partly by debt (instead of selling all \$1,000 equity, it can borrow \$500 and sell only the remaining). If the project's CF is enough to repay the debt, then the **debt is risk-free** (because it does not bear any systematic risk). Equity in a firm that also has debt outstanding is called *levered equity*.
- Since the firm's total CF would still equal the CF of the project, the equity would decrease by the amount of debt. Because the CFs of the debt and equity sum to the cash flows of the project, by the *Law of One Price* the combined values of debt and equity is the firm's value.
- **Law of one price:** *identical investment opportunities trading simultaneously in competitive markets must have the same price!!!* because investors trade rapidly to take advantage of arbitrage opportunities. / The price of a security equals to the PV of its CFs.

TABLE 14.3

Values and Cash Flows for Debt and Equity of the Levered Firm

	Date 0	Date 1: Cash Flows	
	Initial Value	Strong Economy	Weak Economy
Debt	\$500	\$525	\$525
Levered equity	$E = ?$	\$875	\$375
Firm	\$1000	\$1400	\$900

- The *equity is less valuable with leverage*: because the CF of levered equity are smaller than those of unlevered equity, levered equity will sell for a lower price. Since **leverage increases the risk** of the equity (bc debtholders will always be paid before shareholders), it is not appropriate to discount the CF with the same DF (discount factor) used with unlevered equity CFs: *investors in levered equity require a higher expected return to compensate for its increased risk*. Thus, leverage increases the risk of equity even when there is no risk that the firm will default. While debt may be cheaper, it raises the cost of equity, however in the end, the cost of capital of the leveraged and unleveraged firms *will be the same*.

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MMI: no change in value

TABLE 14.4		Returns to Equity with and without Leverage				
	Date 0	Date 1: Cash Flows		Date 1: Returns		
	Initial Value	Strong Economy	Weak Economy	Strong Economy	Weak Economy	Expected Return
Debt	\$500	\$525	\$525	5%	5%	5%
Levered equity	\$500	\$875	\$375	75%	-25%	25%
Unlevered equity	\$1000	\$1400	\$900	40%	-10%	15%

Modigliani-Miller (MM) I

- In a perfect capital market, the total value of a firm's securities is equal to the market value of the total cash flows generated by its assets and is not affected by its choice of capital structure.

OR

In perfect capital markets, the market value of any firm is independent of its capital structure.

- Without taxes and other costs, the *total CFs can be paid out* to the holders of the firm's securities, which is thus equal to CF generated by the assets, so the value of securities and firm's assets should be the same (*Law of One Price*).
- The question may be raised: what if investors would prefer an alternative capital structure (and thus they would like to take less or more risk)?

➔ **Homemade leverage:** if investors want a different capital structure than that of the firm, they can borrow (and add leverage) or lend on their own to achieve these results in their portfolios. If the firm is all-equity financed, the investor can borrow on margin to increase its leverage (and if the CFs of the unlevered equity serve as collateral for the margin loan, then the loan is risk-free). If the equity is leveraged, an investor can replicate the payoffs of unlevered equity by buying both the debt and equity of the firm.

➔ *the entrepreneur's choice of capital structure does not affect the opportunities available to investors*

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TABLE 14.6

Replicating Levered Equity Using Homemade Leverage

	Date 0	Date 1: Cash Flows	
	Initial Cost	Strong Economy	Weak Economy
Unlevered equity	\$1000	\$1400	\$900
Margin loan	-\$500	-\$525	-\$525
Levered equity	\$500	\$875	\$375

- Because the value of a firm is unaffected by its choice of capital structure, *in a perfect capital market*, any financing package (combination of debt and equity) is as good as another, i.e. *any capital structure is optimal*.
- Market value balance sheet:
 - o all assets and liabilities of the firm are included - even intangible assets such as reputation, brand name, or human capital that are missing from a standard accounting balance sheet
 - o all values are current market values rather than historical costs
- The market value balance sheet captures the idea that **value is created by a firm's choice of assets and investments**. By choosing positive-NPV projects the firm can enhance its value.
- **Leveraged recapitalization**: when the company is buying back shares from the proceeds of newly issued debt. The share price should not change because both assets and liabilities increase by the same amount.

TABLE 14.9

Market Value Balance Sheet after Each Stage of Harrison's Leveraged Recapitalization (in \$ million)

Initial		After Borrowing		After Share Repurchase	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
		Cash	Debt	Cash	Debt
		80	80	0	80
Existing assets	Equity	Existing assets	Equity	Existing assets	Equity
200	200	200	200	200	120
200	200	280	280	200	200
Shares outstanding (million)		Shares outstanding (million)		Shares outstanding (million)	
50		50		30	
Value per share		Value per share		Value per share	
\$4.00		\$4.00		\$4.00	

Modigliani-Miller (MM) II

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- Although debt does have a lower cost of capital than equity, we cannot consider this cost in isolation, because it **increases the risk and** therefore the **cost of capital** of the firm's **equity**. MM I states that the total market value of the firm's securities is equal to the market value of its assets, whether the firm is unlevered or levered:

$$E + D = U = A$$

- By holding a portfolio of the firm's E and D, we replicate the CF from holding unlevered equity. The exp. return on debt and equity are called *cost of debt* and *cost of equity*. The overall exp. return is called the *company cost of capital*. In a levered firm, this is:

$$\frac{E}{E + D} R_E + \frac{D}{E + D} R_D = R_U$$

- If we express from this the cost of capital for levered equity, we would find an *additional risk* compared to unlevered equity cost of capital:

$$R_E = \underbrace{R_U}_{\text{Risk without leverage}} + \underbrace{\frac{D}{E}(R_U - R_D)}_{\text{Additional risk due to leverage}}$$

- The amount of **additional risk** depends on the amount of leverage measured by the firm's market value debt-equity ratio, D/E. From this we can conclude MMII:
- *The cost of capital of levered equity increases with the firm's market value debt-equity ratio.*

OR

The exp. rate of return on the common stock of a levered firm increases in proportion to the D/E ratio (expressed in market values).

Cost of Capital of Levered Equity

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

- When the firm was unlevered, equity investors demanded a return of r_A (same as r_U). When the firm is levered, they require a premium $(r_A - r_D) * D/E$ to compensate for the extra risk.

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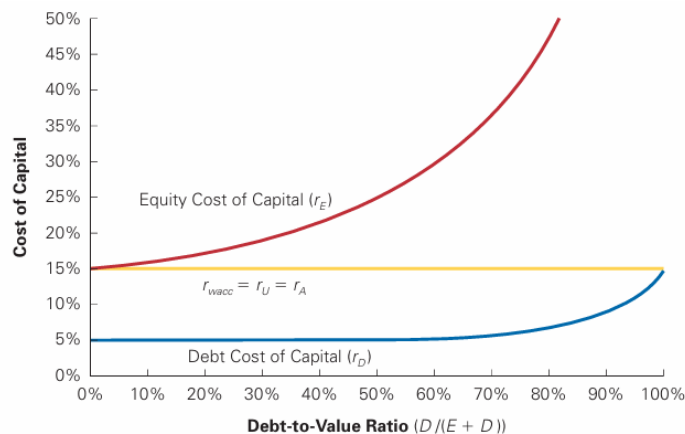
- If a firm is financed by both equity and debt, then the risk of its underlying assets will match the risk of a portfolio of its equity and debt, so the appropriate cost of capital for the firm's assets is the cost of capital of this portfolio, which is simply the weighted average of the firm's equity and debt cost of capital:

$$\begin{aligned}
 &\textbf{Unlevered Cost of Capital (Pretax WACC)} \\
 r_U &\equiv \left(\text{Fraction of Firm Value Financed by Equity} \right) \left(\text{Equity Cost of Capital} \right) + \left(\text{Fraction of Firm Value Financed by Debt} \right) \left(\text{Debt Cost of Capital} \right) \\
 &= \frac{E}{E+D} r_E + \frac{D}{E+D} r_D
 \end{aligned}$$

- Because there are no taxes in perfect capital markets, we do not use the 'normal' WACC, so *with perfect capital markets, a firm's WACC is independent of its capital structure* and is equal to its equity cost of capital if it is unlevered (because all FCF generated by the assets is paid out to the equity holders), which also matches the cost of capital of its assets.

$$r_{wacc} = r_U = r_A$$

- With no debt, the WACC is equal to the unlevered equity cost of capital. As the weight of debt increases, both the equity and the debt become riskier (there is a chance the firm will default) and their cost of capital rises. With 100% debt, the debt would be as risky as the assets themselves (similar to unlevered equity). Yet, because more weight is put on the lower-cost debt, the *weighted average cost of capital remains constant*. (Note, that the r_A is constant only in the MM world, in real life it is a slightly u-shaped curve with a minimum, so there is an *optimal* cost of capital).



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- A firm's *unlevered beta* or *asset beta* is the weighted average of its equity and debt beta. Unlevered beta measures the *market risk of the firm's underlying assets*, and thus can be used to assess the cost of capital for comparable investments.

$$\beta_U = \frac{E}{E+D}\beta_E + \frac{D}{E+D}\beta_D$$

$$\beta_E = \beta_U + \frac{D}{E}(\beta_U - \beta_D)$$

- Cash and risk-free securities can reduce the risk and thus the required risk premium of the firm's assets. They are like negative debt. When we are trying to assess a firm's enterprise value it is natural to measure leverage in terms of the firm's net debt, which is its debt less its holdings of excess cash or short-term investments.

Capital structure fallacies

- *Increasing leverage increases the expected EPS of the firm*, which may look like a good reason to take leverage. However, from MM Proposition I that **as long as the securities are fairly priced, these financial transactions have an NPV of zero** and offer no benefit to shareholders. The 'trick' is that the risk of earning also has to change. When earnings are low, leverage lowers EPS further. Leverage will increase expected EPS whenever the firm's after-tax borrowing cost is less than the ratio of expected earnings to the share price. Even though EPS is higher, due to the additional risk, shareholders will demand a higher return. These effects cancel out, so the *price per share is unchanged*.
- *Issuing equity will dilute* (the cash flows generated by the firm must be divided among a larger number of shares) *existing shareholders' ownership*, so debt financing should be used instead. However, the cash raised by issuing new shares will increase the firm's assets, so as long as the firm sells the new shares of equity at a fair price, there will be no gain or loss to shareholders associated with the equity issue itself (the share price will not change).

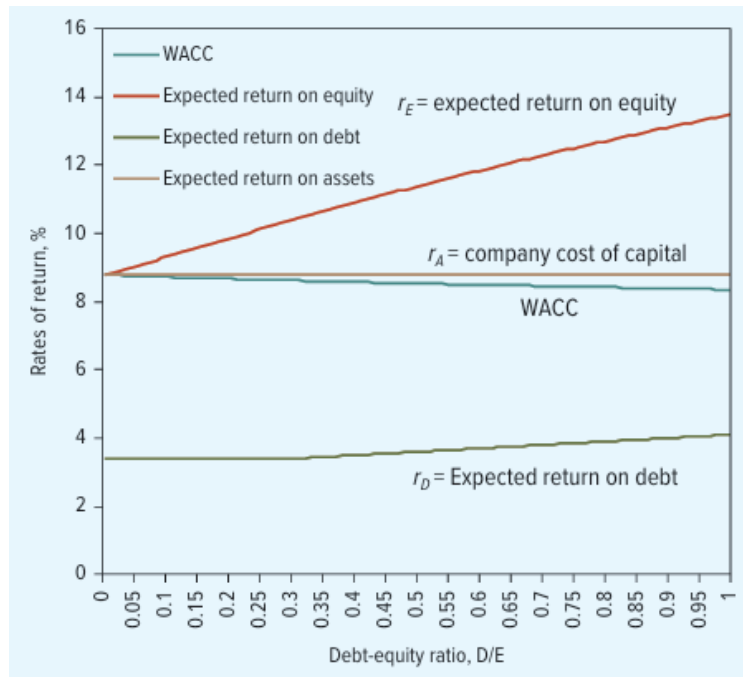
Conservation of value principle for financial markets: *with perfect capital markets, financial transactions neither add nor destroy value, but instead represent a repackaging of risk (and*

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therefore return), i.e., when the firm changes its mix of debt and equity securities, the risk and exp. returns of these securities change, but the company's overall cost of capital does not change.

- In practice, markets are not perfect, and capital structure can have an effect on firm value. MM shows that any effects of capital structure must similarly be due to frictions that exist in capital markets.
- **One complication:** interest paid on a firm's borrowing can be deducted from taxable income
 - o Instead of the company cost of capital, the *after-tax WACC* is used. The after-tax WACC is lower than the opportunity or company cost of capital bc it includes a downward tax adjustment. Contrary to MM II, if companies receive tax shield, the after-tax WACC declines as debt increases (the debt shield will increase).



Weighted Average Cost of Capital (After Tax)⁶

$$r_{wacc} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - \tau_c)$$

Debt and taxes

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- In *perfect capital markets* the law of one price implies that all financial transactions have a NPV of zero and neither create nor destroy value, that's why firm value is independent of capital structure. But if capital structure does matter, then it means that there are **market imperfections**. One such imperfection is the presence of **taxes**.
- Corporations must pay income taxes, but interest expenses reduce this amount. This gives an incentive to use debt. Debt obligations reduce the income available to *equity holders*. However, the total amount available to *all investors* is higher with leverage. The company below pays less tax with leverage, but it has less Net Income too. However, it pays out **1,960** (1,560+400) to *all investors*, while without leverage it would pay out only **1,820** to *shareholders*. The 140 difference comes from the reduction in taxes (840 vs 980).

	With Leverage	Without Leverage
EBIT	\$2800	\$2800
Interest expense	-400	0
Income before tax	2400	2800
Taxes (35%)	-840	-980
Net income	\$1560	\$1820

- The gain to investors from the tax deductibility of interest payments is referred to as the **interest tax shield (ITS)**.

$$\text{Interest Tax Shield} = \text{Corporate Tax Rate} \times \text{Interest Payments}$$

- The ITS provides benefit each year so we can value it by calculating the PV of this CF

$$\left(\begin{array}{c} \text{Cash Flows to Investors} \\ \text{with Leverage} \end{array} \right) = \left(\begin{array}{c} \text{Cash Flows to Investors} \\ \text{without Leverage} \end{array} \right) + (\text{Interest Tax Shield})$$

- By increasing the amount paid to debt holders through interest payments, the amount of the pretax cash flows that must be paid as taxes decreases. The gain in total cash flows to investors is the ITS.
- Change to MM I: **The total value of the levered firm exceeds the value of the firm without leverage due to the present value of the tax savings from debt:**

$$V^L = V^U + PV(\text{Interest Tax Shield})$$

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- So, when a firm uses debt financing, the cost of the interest it must pay is offset to some extent by the tax savings from the interest tax shield → *with tax-deductible interest, the effective after-tax borrowing rate is $r(1 - \tau_c)$.*
- The WACC represents the effective cost of capital to the firm, after including the benefits of the interest tax shield. It is therefore lower than the pretax WACC. WACC is only valid when TS in any period is just equal to the market value of debt * cost of debt ($D * r_D$). Once there is some return on debt that is not tax deductible for the borrower, then this is already not true. It is already assumed that there is no TS effect on financing cost linked to equity.

Weighted Average Cost of Capital (After Tax)⁶

$$r_{wacc} = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D (1 - \tau_c)$$

- **Personal taxes:** by reducing a firm's corporate tax liability, debt allows the firm to pay more of its cash flows to investors. Unfortunately for investors, after they receive the cash flows, they are generally taxed again. Personal taxes have the potential to offset some of the corporate tax benefits of leverage, so the combined effect of both corporate and personal taxes has to be evaluated. After this evaluation, the effective tax advantage of debt is:

Effective Tax Advantage of Debt

$$\tau^* = 1 - \frac{(1 - \tau_c)(1 - \tau_e)}{(1 - \tau_i)}$$

- Taxes change the **optimal capital structure**. To receive the full tax benefits of leverage, a firm does not need to use 100% debt financing. Moreover, no corporate tax benefit arises from incurring interest payments that regularly exceed EBIT. Therefore, *the optimal level of leverage from a tax saving perspective is the level such that interest equals EBIT.*

TABLE 15.4

Tax Savings with Different Amounts of Leverage

	No Leverage	High Leverage	Excess Leverage
EBIT	\$1000	\$1000	\$1000
Interest expense	0	-1000	-1100
Income before tax	1000	0	0
Taxes (35%)	-350	0	0
Net income	650	0	-100
Tax savings from leverage	\$0	\$350	\$350

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- However, it is unlikely that a firm can predict its future EBIT precisely, so it is wise to reduce the level of interest payment, to not exceed the actual EBIT in the future. To avoid excess interest, the firm should have debt with interest payments that are below its expected taxable earnings (EBIT). The optimal fraction of debt and thus the optimal leverage ratio is also affected by *growth* of earnings. The higher the growth rate, the higher the value of the equity. As a result, *the optimal proportion of debt in the firm's capital structure $[D/(E + D)]$ will be lower, the higher the firm's growth rate.*
- In real life companies do not reach this optimal level, since debt has other costs too: increasing the level of debt increases the *probability of bankruptcy*.
- Although debt has tax advantages over equity financing, too much debt results in high leverage, *leading to high interest rates* to compensate lenders for the increased risk of default.